



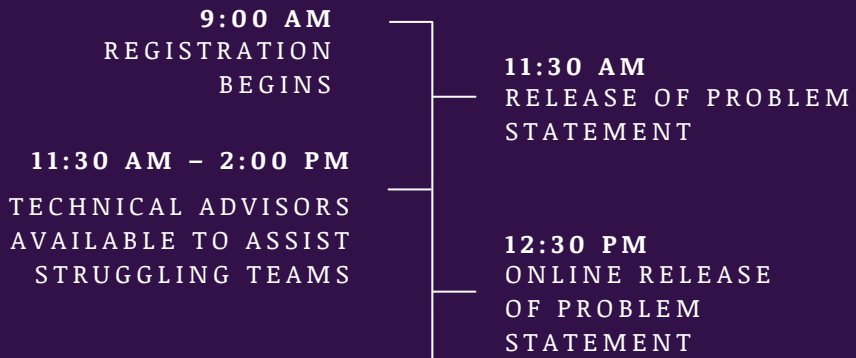
Datathon Handbook

3rd -4th OCTOBER 2026

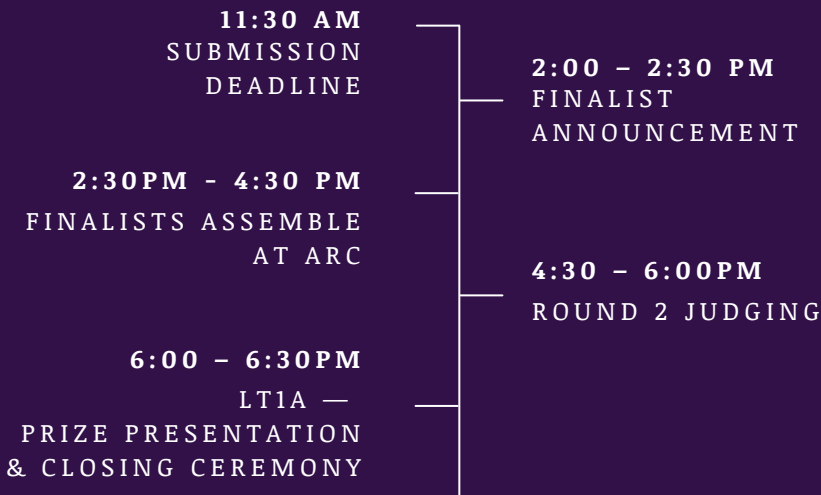
TIMELINE

Below is the detailed timeline of our event, so you know what to expect at every stage.

OCT 3



OCT 4



WELCOME NOTE

EMPOWERING THE NEXT GENERATION OF PROBLEM SOLVERS

Are you ready to turn data into meaningful solutions? Whether you're passionate about sustainability, interested in finance, or simply looking to challenge yourself, this datathon is an opportunity to explore real-world problems through data.

Across 3–4 October 2026, you will work in teams to tackle challenges. Along the way, you'll have access to workshops and technical guidance to help you develop your ideas, strengthen your analytical skills, and turn data into actionable insights.

Whether you're new to data analytics or already familiar with machine learning, come ready to learn, collaborate, and solve with us

Best Regards,
Deep Learning Week Committee





OVERVIEW

Round 1

- The Problem Statements will be released during the Opening Ceremony at 11:30AM on 3 October. (Online at 12:30 PM)
- Teams will work on one of two tracks:
 - Track 1 – Sustainability,
 - Track 2 – Finance
- Teams will analyse the provided data and train the model according to the problem statement
- Latest submission: **11:30 AM, 4 October 2026.**
- Finalists will be selected based on the first-round submission and announced between **2:00 PM – 2:30 PM on 4 October.**

Part 2: Final Round

- **Venue:** ARC
 - **Registration:** Opens at 3:30 PM, 4 October
 - **Format:** Each finalist team will be assigned a booth for a showcase format where judges and other finalists will view your work.
 - **Presentation:** Teams will present their solutions directly from their Jupyter Notebook (slides are optional). Be prepared for a brief demo and Q&A with the judges.
 - **What to Bring:** Please bring your own laptop, charger and HDMI adaptor for display at your booth.
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ROUND 1 DELIVERABLES

Note – All instructions provided applies to both tracks

Each team is required to submit the following through the official submission portal. Please coordinate with your teammates, as only a single submission per team will be accepted.

1. A **single Jupyter Notebook file** (.ipynb)
2. Trained model (.pkl) - look at instructions below to download the model
3. A **1-page** technical report (.pdf)
4. Requirement file (.txt)

The required training and test datasets will be provided to you. You can however, curate your own datasets or use publicly available ones if you feel they would help improve your model's scores.

Jupyter Notebook File

Your notebook should contain your complete machine-learning solution, including but not limited to:

- Data loading and preprocessing
- Exploratory data analysis
- Feature engineering
- Model training
- Model evaluation
- Prediction generation
- Any other steps required to reproduce your solution

Technical Proposal

Your one-page technical proposal should briefly explain your understanding of the problem, how you prepared and explored the data, and the approach you took to develop your solution. Focus on the reasoning behind your decisions, key insights from your work, and any limitations or improvements you would consider.

UPLOAD INSTRUCTIONS

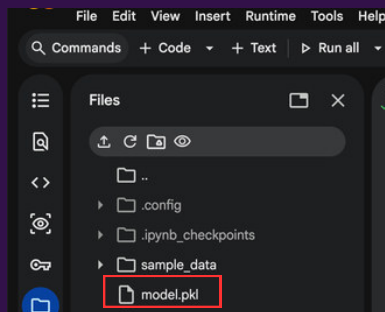
Step 1 - After training your model; follow the steps below to download your model as a .pkl file

```
# 1. Train your model
X_train = make_features(train_df)
feature_names = ["col_a", "col_b", "feature_ratio"]

model = RandomForestRegressor(random_state=42)
model.fit(X_train[feature_names], train_df["target"])

# 2. Save model AND feature names list into model.pkl
import joblib
joblib.dump((model, feature_names), "model.pkl")
```

Step 2 - In the left-pane of Google Colab, click on the files icon. You should be able to see model.pkl file you just downloaded



Step 3 - Next in a new Google Colab file, do the following steps

- Copy the feature engineering you did
- Load the model.pkl file into the new Google Colab
- Use the model to predict and save the output to csv in the required format

```
# 1. Feature engineering (must match training exactly)
def make_features(df):
    df = df.copy()
    df["feature_ratio"] = df["col_a"] / (df["col_b"] + 1e-5)
    return df

# 2. Load trained model and saved feature list
model, features = joblib.load("model.pkl")

# 3. Read test features (uses environment variable on sandbox, test.csv locally)
input_path = os.environ.get("DATATHON_INPUT_PATH", "test.csv")
test_df = pd.read_csv(input_path)

# 4. Apply feature engineering & predict
X_test = make_features(test_df)
preds = model.predict(X_test[features])

# 5. Save predictions
pd.DataFrame({"prediction": preds}).to_csv("BEST_TEAM_Datathon_2026_Track_1_Prediction", index=False)
print("Predictions saved successfully!")
```

LEADERBOARD

Public

Upload your prediction on the test dataset to view your score. You can track your progress and compare across different models too.

! Important - This is only an indication of performance during the datathon.

Private

After the datathon ends, we will be running your submitted model against a hidden private test dataset.

Private evaluation may contain different observations and edge cases from the public test data.

Your public leaderboard rank does not necessarily reflect your final private leaderboard rank.

SUBMISSIONS

Only a single copy of your work will be accepted through the official portal ([Click here](#)). Please coordinate with your teammates. The login details will be sent to the leader of your team via email.

Before submitting, you must run your notebook from start to finish in a clean environment such as Google Colab to verify that it executes without errors, contains all required imports and dependencies, and produces the correct output format and dimensions.

Any execution error during evaluation may result in disqualification.

Deadline: 11:30AM, 4th October 2026

PROBLEM STATEMENT

Track 1 - Smart Campus Analytics

CONTEXT

Universities operate a wide range of facilities, each with different patterns of energy consumption. These patterns can be influenced by factors such as time of day, occupancy, weather, and previous energy usage.

PROBLEM STATEMENT

How accurately can we predict the energy consumption of a campus facility using historical usage patterns, environmental conditions, and building information?

WHAT ARE WE LOOKING FOR?

You will receive a training dataset and a testing dataset. Your goal is to train a model of your choice to predict the energy consumption.

You are expected to work through this problem using the machine learning pipeline

1. Understanding the problem
2. Explore the Dataset
3. Clean and preprocess the data
4. Engineer useful features
5. Build a baseline model
6. Train & Compare models
7. Evaluate performance

Advice - better modelling is not necessarily about using the most complicated model. It is about selecting the right model for the right situation.

EVALUATION

The evaluation process takes place over 2 rounds. Submissions are evaluated based on the following criteria for the first round. Thereafter, shortlisted teams will proceed to the final stage.

Round #1:

- **60%** – The main metric we will be assessing is the Root Mean Squared Error (RMSE), MAE and R-square.
- **40%** – We will also be assessing the team's understanding of the problem statement, why the specific model is used and what are the assumptions of the model?

Final Round: Presentations will be conducted using your submitted Jupyter Notebook; slides are optional. Judges will assess your technical understanding of the problem, how effectively you translate that into modeling, and your presentation skills. Please be prepared for a Q&A session with the judges.

PRIZES

Note that each one in your team will have the following item

1st Place - Armaggeddon keyboard

2nd Place - Logitech Mouse

3rd Place - Jisulife Fan

Track 2 - Intelligent Financial Fraud Detection

CONTEXT

Financial institutions process large numbers of transactions, but only a small proportion may be fraudulent. This creates a challenging machine learning problem: a model must identify rare fraudulent transactions without incorrectly flagging too many legitimate ones.

The challenge is therefore not simply to maximise accuracy, but to develop a model that can effectively distinguish fraudulent transactions from legitimate ones.

PROBLEM STATEMENT

How can machine learning identify potentially fraudulent financial transactions while minimising unnecessary disruption to legitimate customers?

WHAT ARE WE LOOKING FOR?

You will receive a labelled training dataset and an unlabelled test dataset. Your goal is to train a model of your choice to predict whether the transaction is a fraud or not (label) from the test dataset.

You are expected to work through this problem using the machine learning pipeline

1. Understanding the problem
2. Explore the Dataset
3. Clean and preprocess the data
4. Engineer useful features
5. Build a baseline model
6. Train & Compare models
7. Evaluate performance

EVALUATION

The evaluation process takes place over 2 rounds. Submissions are evaluated based on the following criteria for the first round. Thereafter, shortlisted teams will proceed to the final stage.

Round #1:

- **60%** – The main metric we will be assessing are the PR-AUC, F1 and Recall
- **40%** – We will also be assessing the team's understanding of the problem statement, why the specific model is used and what are the assumptions of the model?

Final Round: Presentations will be conducted using your submitted Jupyter Notebook; slides are optional. Judges will assess your technical understanding of the problem, how effectively you translate that into modeling, and your presentation skills. Please be prepared for a Q&A session with the judges.

PRIZES

Note that each person in your team will have the following item

1st Place - Acer Monitor 27"

2nd Place - Sony Wireless Earbuds

3rd Place - Power bank

FAQs

1. How does the elimination round work?

We will run your model against a private test dataset and generate a score using the metrics above.. On top of this, we will read through your Notebook to determine whether you truly understand what the model is doing.

2. What should I bring to the event?

Bring your laptop, your creativity and critical thinking :)

3. Will food be provided during the event?

Tea break will be provided for finalists on 4th October

4. How do we submit our solutions for the final round?

No additional submission required. You will be primarily using the submitted Jupyter Notebook to present to the judges physically in NTU. You'll have time to touch up your notebook before the judging round starts.

5. What are the judges looking for during the presentations?

Judges will evaluate how thoroughly you understand the dataset, how effectively you translate those insights into model building and training, and how clearly you present your work.

For more information relating to the event please visit our website
dlweek.com

CONTACTS

Connect with us on all our socials!



If you have further questions, feel free to contact:

deeplearningweek@gmail.com